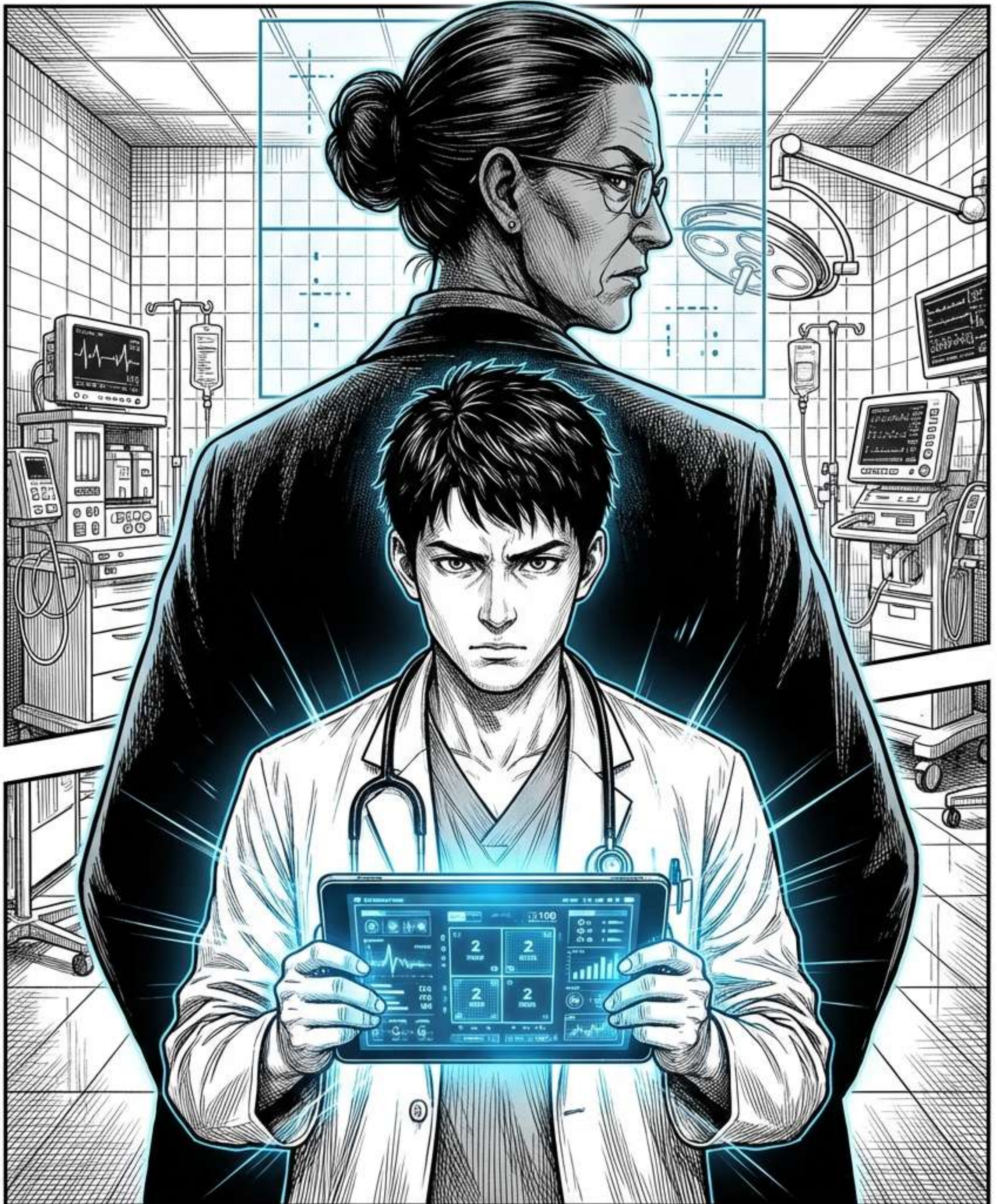
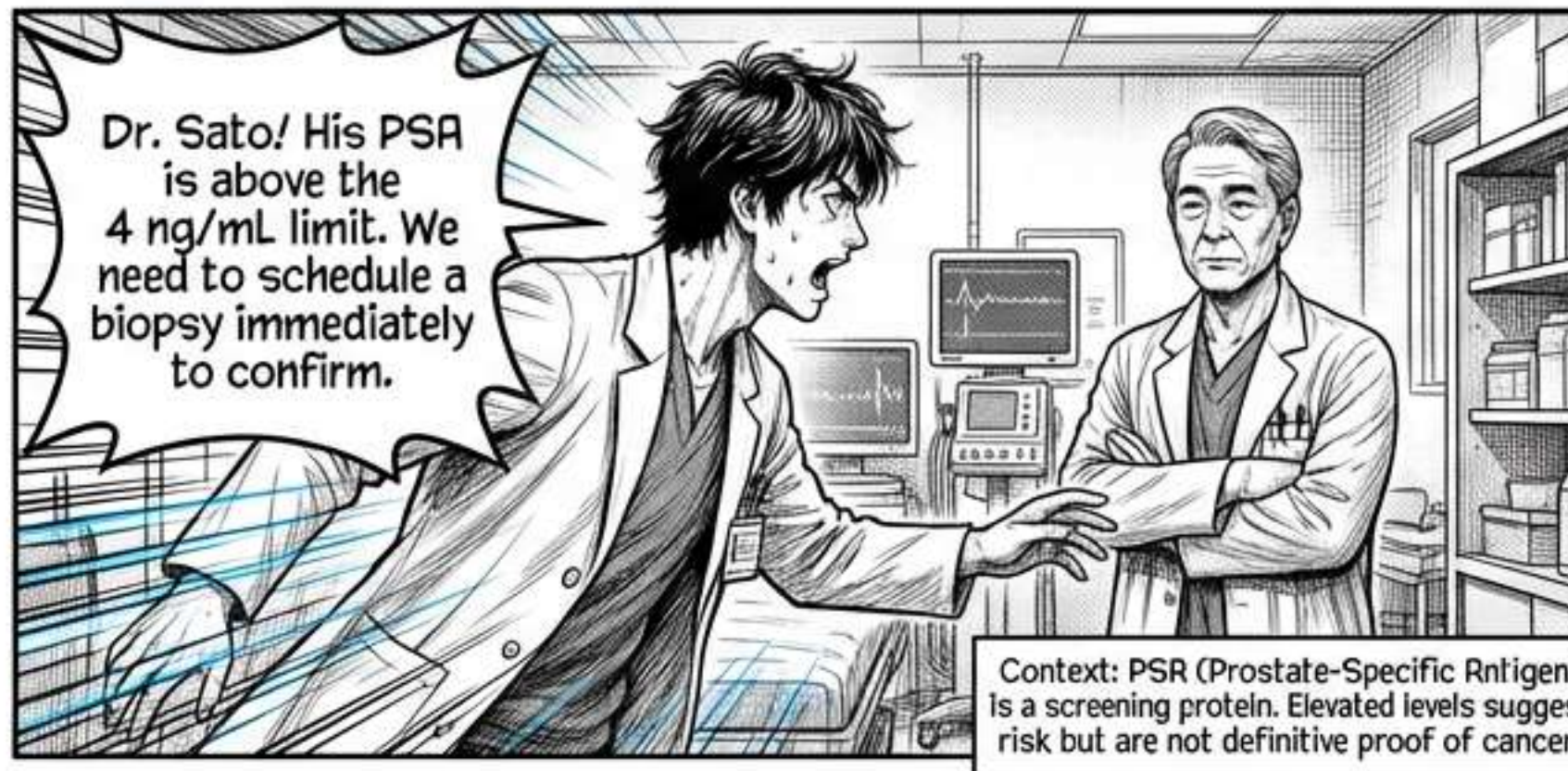


THE DIAGNOSTIC THRESHOLD

A Statistical Case Study in Sensitivity, Specificity, and Predictive Values



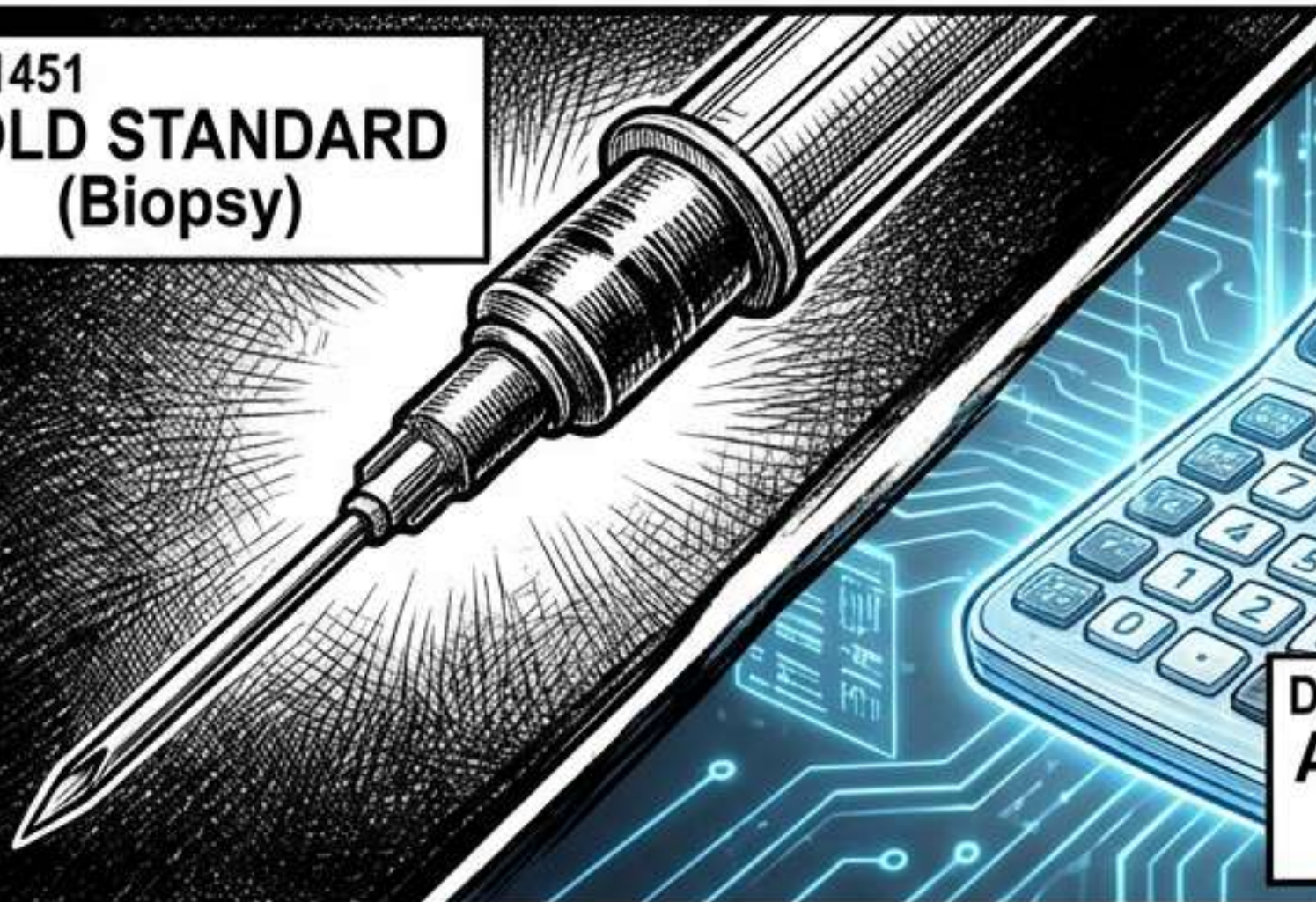
Based on "Foundational Statistical Principles in Medical Research" by Monaghan et al.  NotebookLM





Pause, Kenji.
A biopsy is the
"Gold Standard"—
it tells the absolute
truth—but it is
invasive, painful,
and costly.

DIN 1451
GOLD STANDARD
(Biopsy)



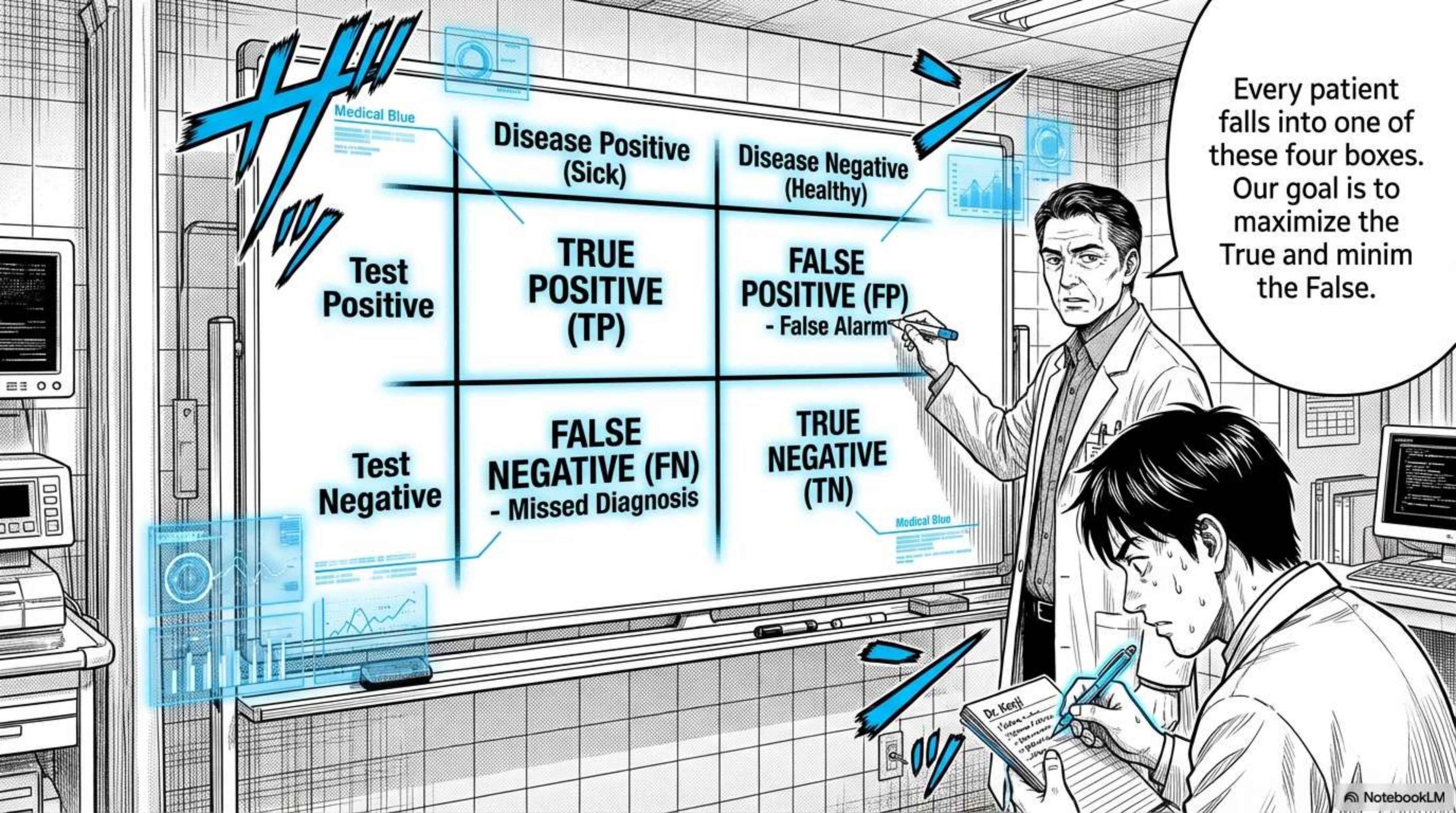
DIN 1451
ALTERNATIVE TEST
(PSA Density)

We must balance
yield with pragmatism.
We will calculate the
PSA Density (PSAD).
We use the study cutoff
of 0.08 ng/mL/cc.

DIN 1451

Study Cutoff:
0.08 ng/mL/cc





Every patient falls into one of these four boxes. Our goal is to maximize the True and minim the False.

Disease Positive (Sick)

Disease Negative (Healthy)

Test Positive

TRUE POSITIVE (TP)

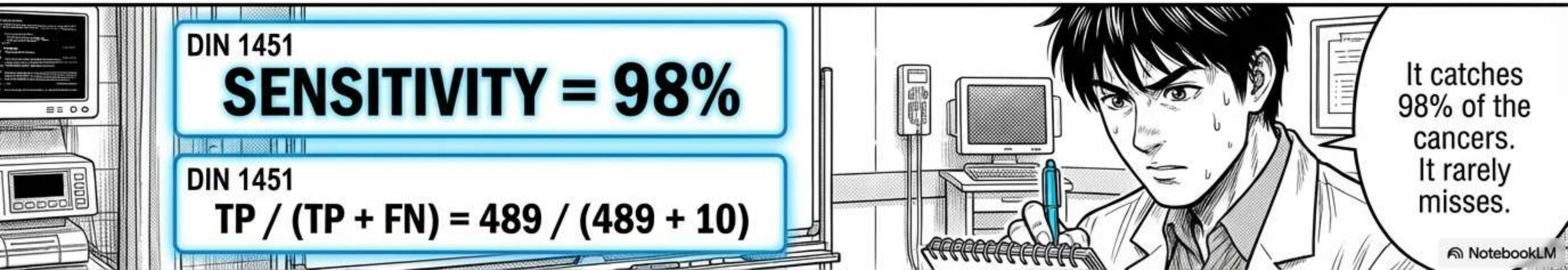
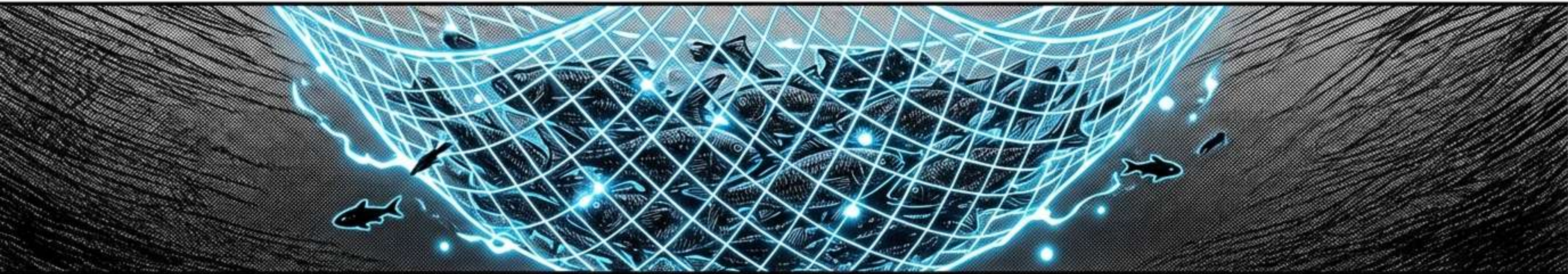
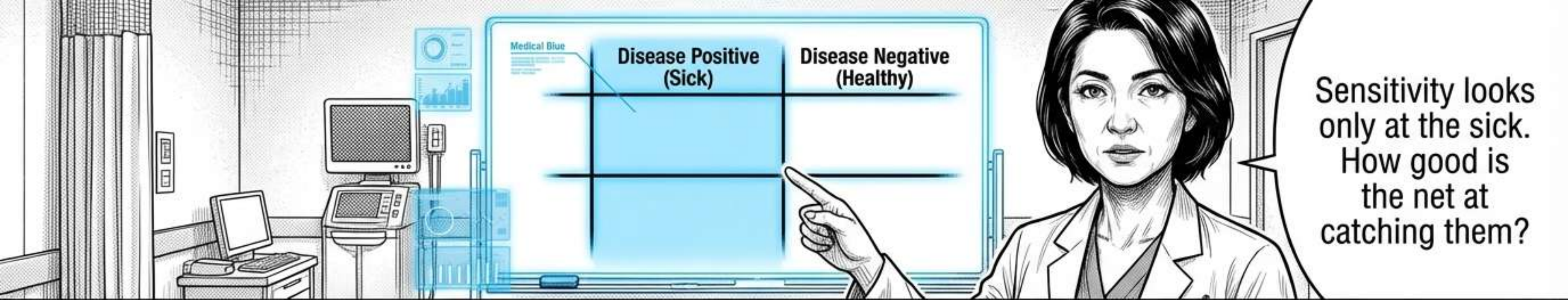
FALSE POSITIVE (FP)
- False Alarm

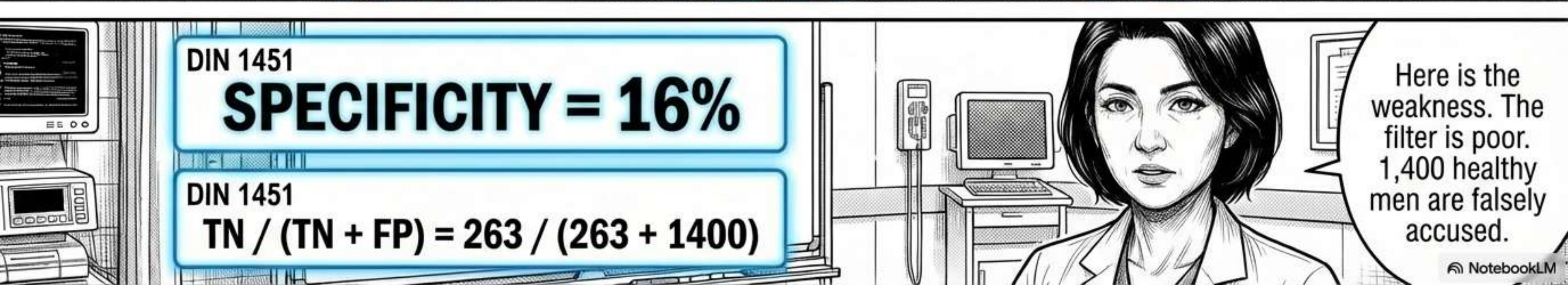
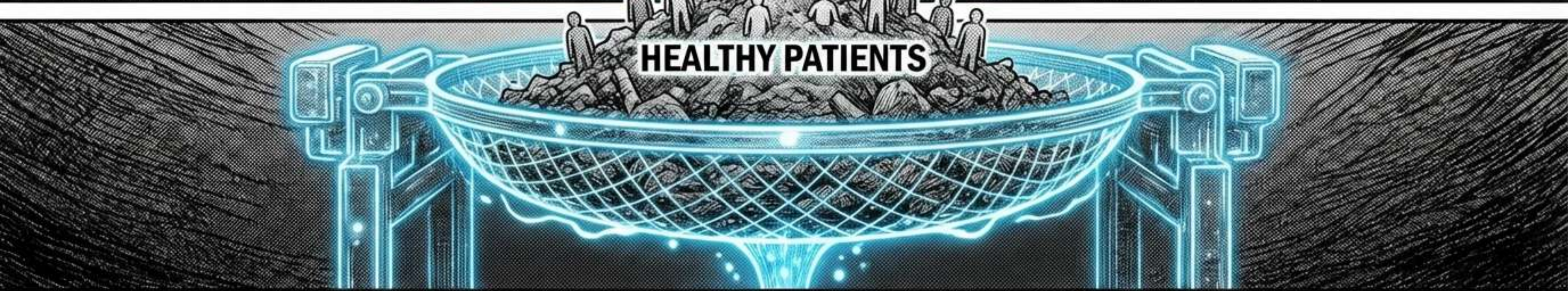
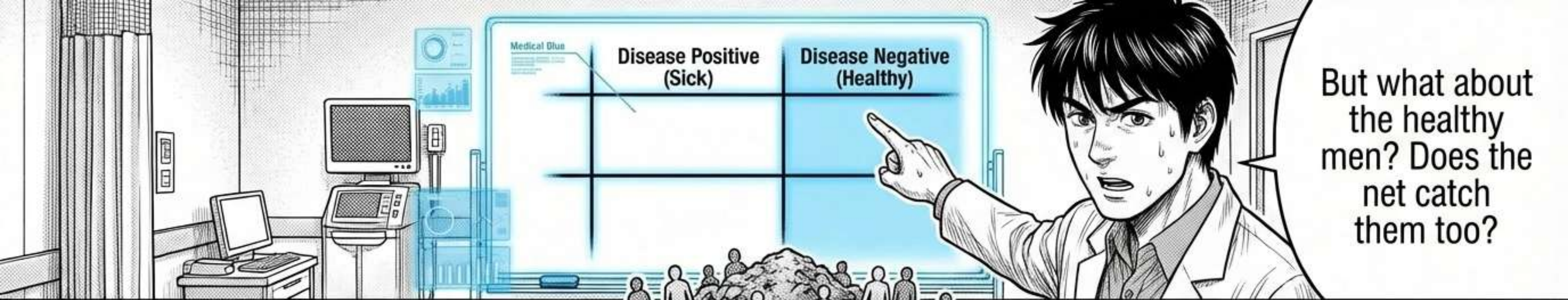
Test Negative

FALSE NEGATIVE (FN)
- Missed Diagnosis

TRUE NEGATIVE (TN)

Dr. Kenji
I have a patient
who is
very
sick
and
I
need
to
see
you
today.





THRESHOLD (PSAD)

0.08



0.08



Cutoff 0.05

DIN 1451
Catch Everyone
False Alarm Everyone.

99.6%



SENSITIVITY

3%

SPECIFICITY

See-Saw

57%

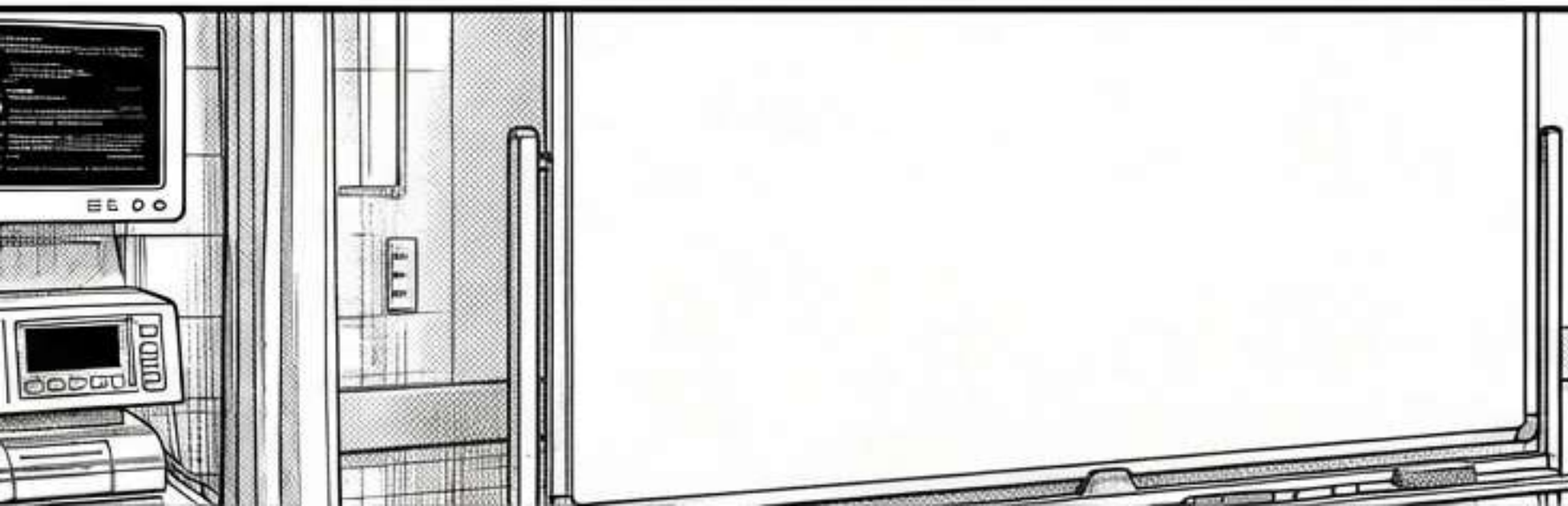
SPECIFICITY

72%

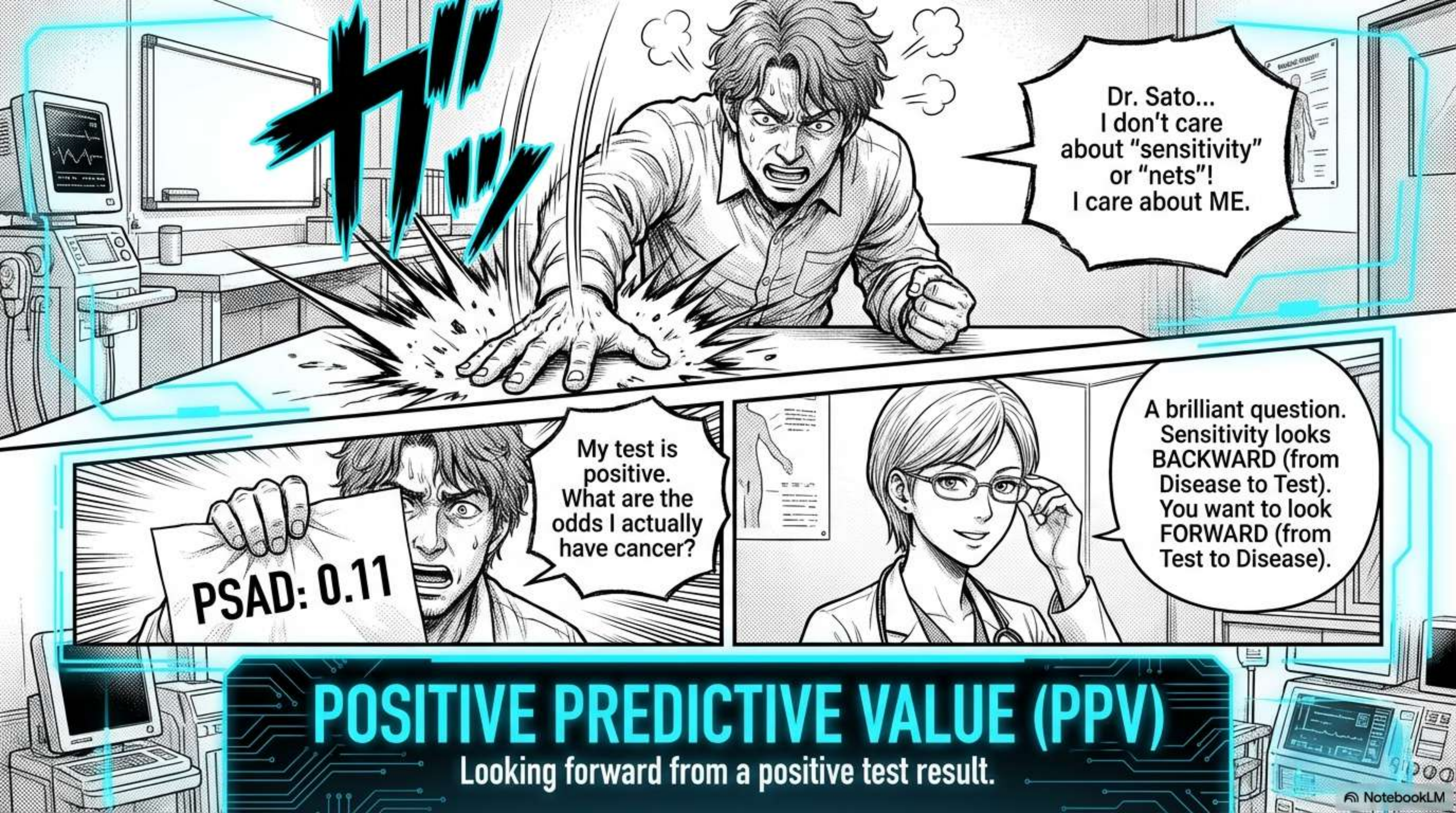
SENSITIVITY

Cutoff 0.15

DIN 1451
Better Filter
Missed Cancers.



You cannot maximize both. If you tighten the net to reduce false alarms, you will let cancer slip through.



Dr. Sato...
I don't care
about "sensitivity"
or "nets"!
I care about ME.

My test is
positive.
What are the
odds I actually
have cancer?

PSAD: 0.11

A brilliant question.
Sensitivity looks
BACKWARD (from
Disease to Test).
You want to look
FORWARD (from
Test to Disease).

POSITIVE PREDICTIVE VALUE (PPV)

Looking forward from a positive test result.

PPV CALCULATION



ALL POSITIVE TESTS



PPV VISUALIZATION

$$\text{PPV} = \frac{\text{True Positives}}{\text{All Positives}}$$

$$\frac{489}{(489 + 1400)} = 26\%$$



Mr. Tanaka, despite the positive test, there is only a 26% chance you actually have the disease.

Looking forward from a positive test result.



Test Result:
Negative (< 0.08)

ALL NEGATIVE TESTS



True Negatives
(263)

False Negatives
(10)

NEGATIVE UTILIZATION

$$\text{NPV} = \frac{\text{True Negatives}}{\text{All Negatives}}$$

$$\frac{263}{(263 + 10)} = 96\%$$



If the test was negative, we could be 96% sure you were healthy. This test is better for RULING OUT disease.

Looking forward from a positive test result.

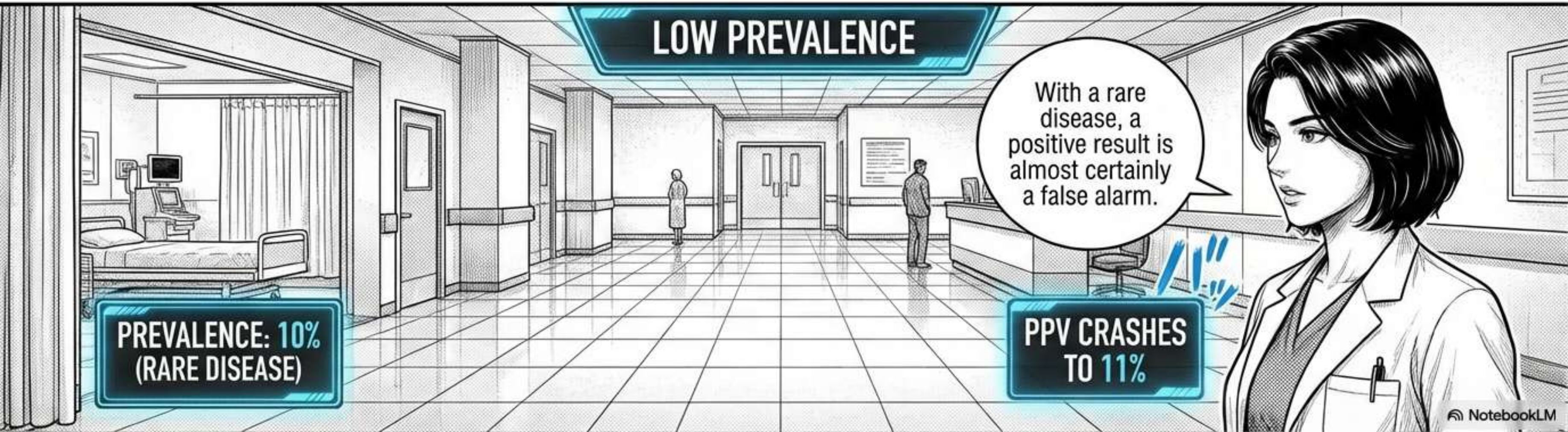
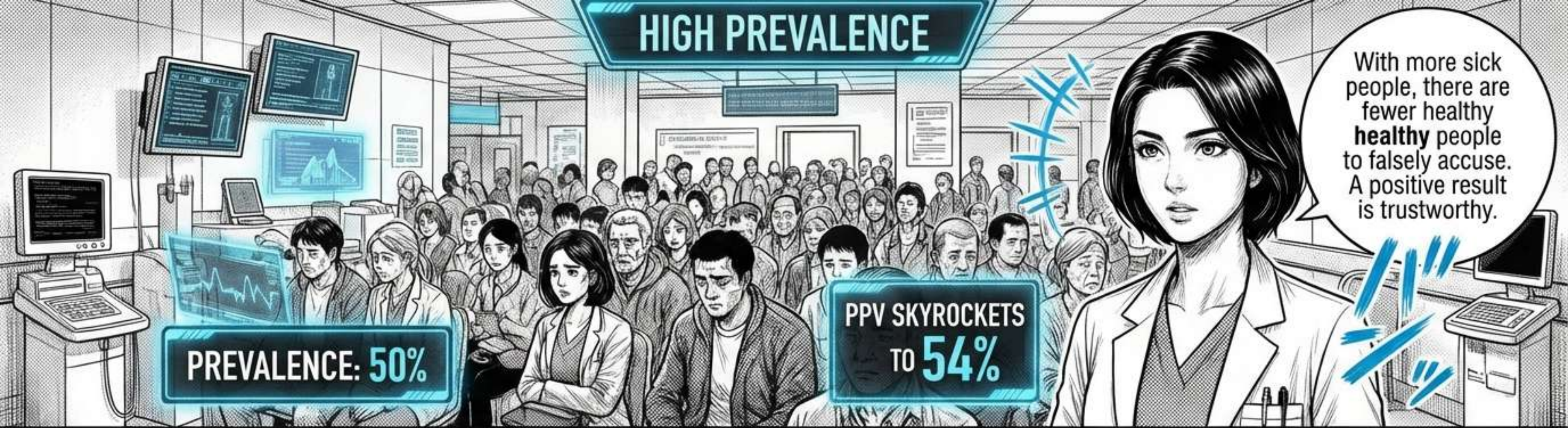
THE PREVALENCE FACTOR

SPECIALIZED
CANCER CENTER

Now, imagine a twist.
We move to a hospital
where 50% of the men
have cancer, instead of
just 23%.

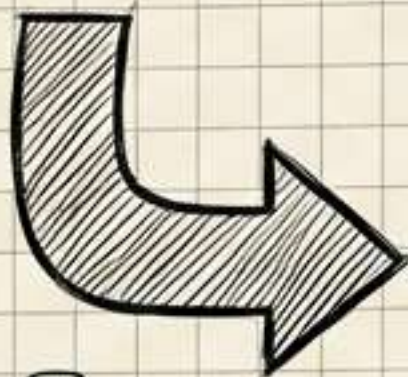
Sensitivity and Specificity are
STABLE (Test properties).

PPV and NPV are **UNSTABLE**
(Depend on Prevalence).

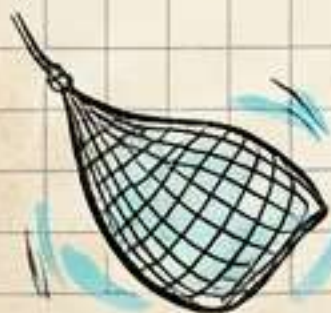


STABLE (The Machine)

- Sensitivity
- Specificity



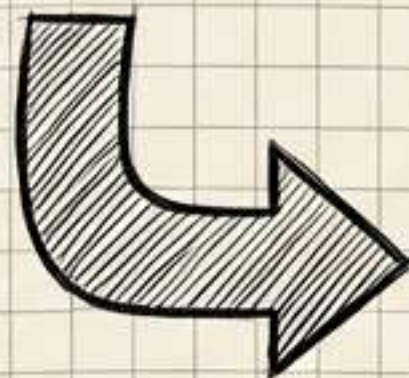
Defined by the
test cutoff.
Unaffected by
population.



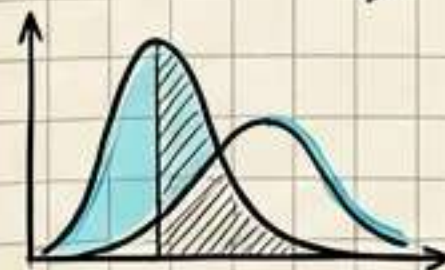
*"Sensitivity is the net.
number of*

UNSTABLE (The People)

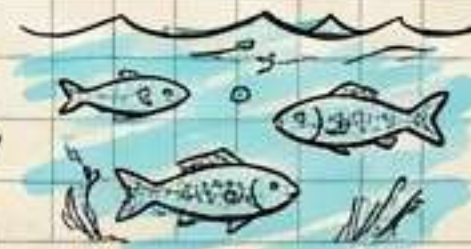
- PPV
- NPV



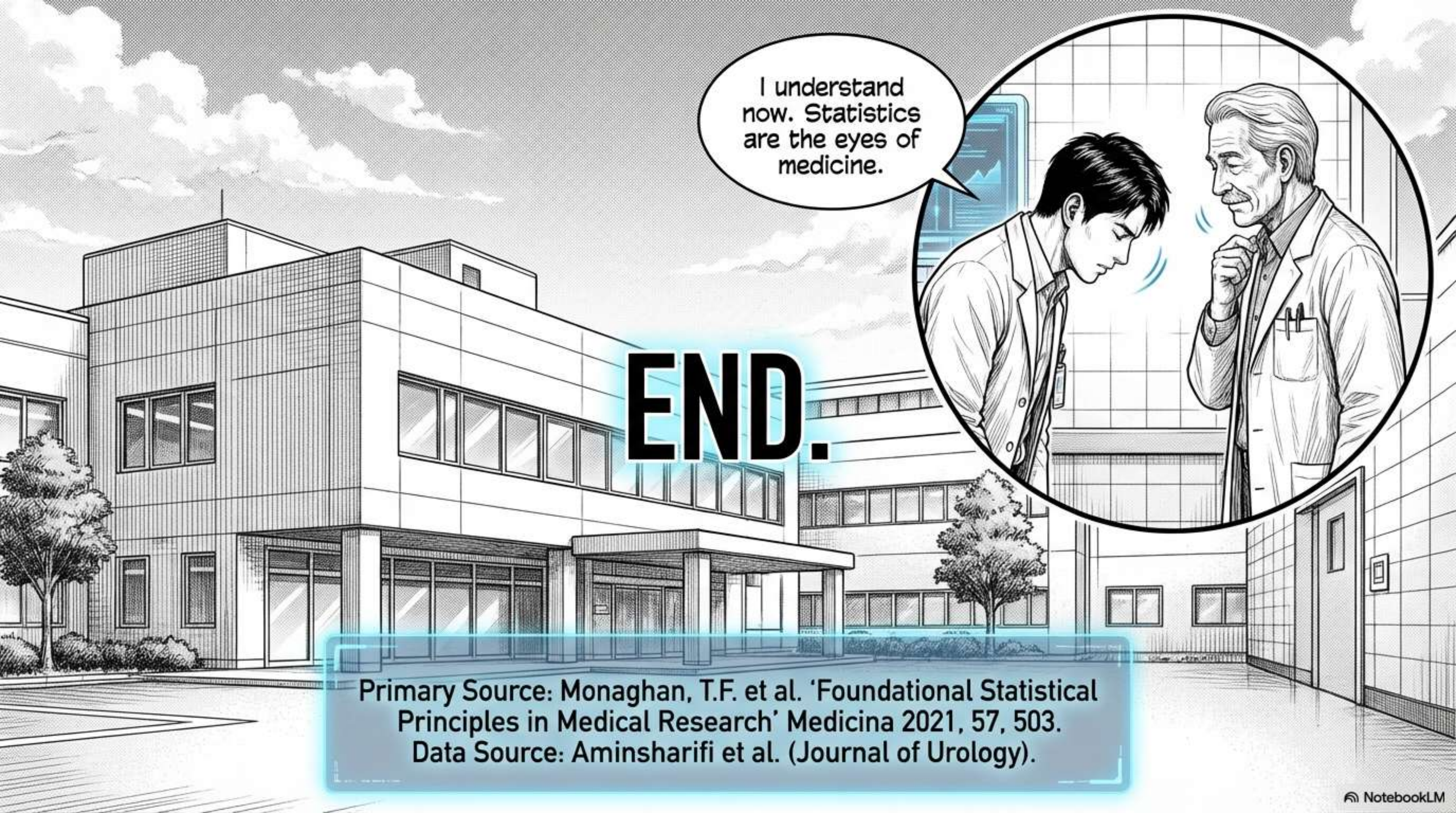
Defined by
Prevalence
(Pre-test
probability).



*Prevalence is the
fish in the water."*







I understand now. Statistics are the eyes of medicine.

END.

Primary Source: Monaghan, T.F. et al. 'Foundational Statistical Principles in Medical Research' Medicina 2021, 57, 503.
Data Source: Aminsharifi et al. (Journal of Urology).